

생명과학개론 (2024년 1학기 기말고사)

<영어단어 각2점 / 총30점>

1. An RNA molecule that functions as an enzyme. **Ribozyme**
2. The community of microorganisms that live in and on the body of an animal. **Microbiota**
3. The transformation of a larva into an adult. **Metamorphosis**
4. The densely branched network of hyphae in a fungus. **Mycelium**
5. The evolution of similar features in different evolutionary lineages, which can result from living in very similar environments. **Convergent evolution**
6. The meshwork that surrounds animal cells, consisting of a web of protein and polysaccharide fibers embedded in a liquid, jelly, or solid. **Extracellular matrix**
7. The first portion of the vertebrate small intestine after the stomach, where chyme from the stomach is mixed with bile and digestive enzymes. **Duodenum**
8. The steady state of body functioning; the tendency to maintain relatively constant conditions in the internal environment even when the external environment changes. **Homeostasis**
9. Rhythmic waves of contraction of smooth muscles. This propels food through a digestive tube and also enables many animals, such as earthworms, to crawl. **Peristalsis**
10. A cardiovascular disease in which a buildup of fatty deposits called plaque develops on the inner walls of the arteries, narrowing the passageways through which blood can flow. **Atherosclerosis**
11. A long-lived lymphocyte that responds to subsequent exposures to a specific pathogen. These cells are formed during the primary immune response and is activated by exposure to the same antigen that triggered its formation. **Memory cells**
12. The main control center of the endocrine system, located in the vertebrate brain. This functions in maintaining homeostasis, especially in coordinating the endocrine and nervous systems. **Hypothalamus**
13. A junction, or relay point, between two neurons or between a neuron and a muscle or gland cell. Electrical and chemical signals are relayed from one cell to another at this. **Synapse**
14. The body's main system for internal chemical regulation, consisting of all hormone-secreting cells; cooperates with the nervous system in regulating body functions and maintaining homeostasis. **Endocrine system**
15. A group of populations the members of which have the potential to interbreed and produce viable, fertile offspring.

Species

<단답식 각3점 / 총30점>

1. What process is the ultimate source of the genetic variation that serves as raw materials for evolution?

Mutation

2. A paleontologist estimates that when a particular rock formed, it contained 12 mg of the radioactive isotope potassium-40. The rock now contains 3 mg of potassium-40. The half-life of potassium-40 is 1.3 billion years. About how old is the rock? **2.6 billion years**

3. Place these events in the history of life on Earth in the order that they occurred.

- (1) accumulation of O₂ in Earth's atmosphere
- (2) colonization of land by plants and fungi
- (3) diversification of animals (Cambrian explosion)
- (4) origin of eukaryotes
- (5) origin of humans
- (6) origin of multicellular organisms
- (7) origin of prokaryotes

정답 7-1-4-6-3-2-5

4. A neuron ends in many branches, each with a bulb-like _____ that contains neurotransmitters, which can be used to communicate signals to other neurons.

- (1) synaptic terminal
- (2) axon
- (3) dendrite
- (4) cell body

정답 (1)

5. You discover a new species of plant. Under the microscope, you find that it produces flagellated sperm. A genetic analysis shows that its dominant generation has diploid cells. What kind of plant do you have?

정답 Fern (양치식물) 또는 고사리(류)

6. Match each type of defensive cells with its function or description.

(1) lymphocyte	(A) attacks infected body cells
(2) cytotoxic T cell	(B) secretes antibodies
(3) helper T cell	(C) white blood cell that engulfs cells or molecules
(4) phagocytic cell	(D) general name for B or T cell
(5) B cell	(E) initiates the secondary immune response
(6) memory cell	(F) cell most commonly attacked by HIV

정답 1(D), 2(A), 3(F), 4(C), 5(B), 6(E)

7. Match each chamber of the heart with its function.

(1) left atrium	(A) receives blood from the body via the inferior venae cavae
(2) right atrium	(B) sends blood to the lungs via the pulmonary arteries
(3) left ventricle	(C) receives blood from the lungs via the pulmonary veins
(4) right atrium	(D) sends blood to the body via the aorta

(정답) 1C, 2A, 3D, 4B

8. Which of the following controls the activity of all the others?

- (1) hypothalamus
- (2) thyroid gland
- (3) pituary gland
- (4) adrenal cortex

정답 (1)

9. Which of the following best illustrates homeostasis?

- (1) Most adult human beings are between 5 and 6 feet tall.
- (2) The lungs and intestines have large surface areas for exchange.
- (3) When the concentration of water in the blood goes up, the kidneys expel more water
- (4) When oxygen in the blood decreases, you may feel light-headed.

정답 (3)

10. Reptiles are much more extensively adapted to life on land than amphibians because reptiles _____.

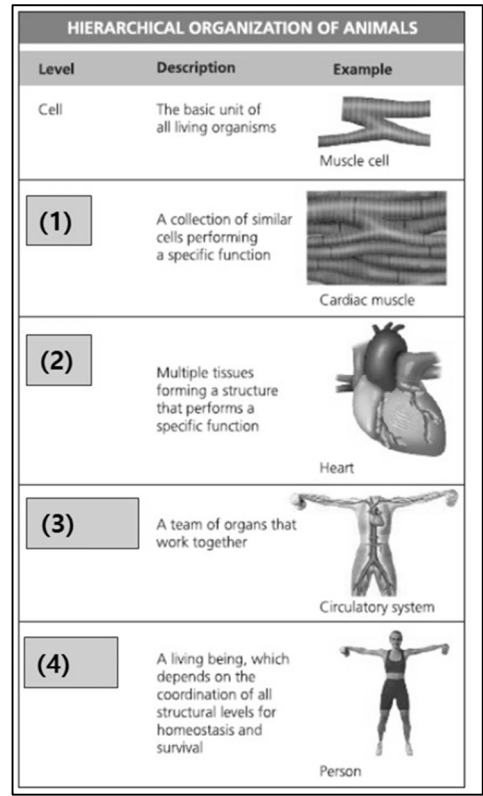
- (1) have a complete digestive tract
- (2) lay eggs that are enclosed in shells
- (3) are endothermic
- (4) go through a larval stage

정답 (2)

<서술형 각5점 / 총40점>

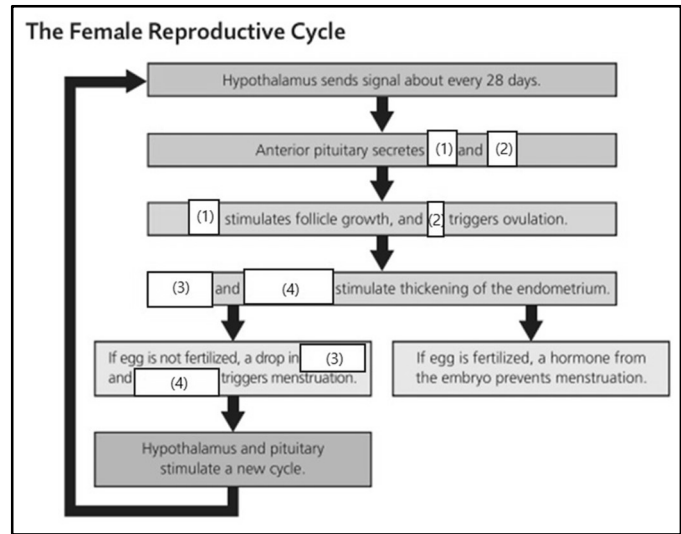
1. 다음은 동물이 구조적으로 구성되어 있는 단계를 나타낸 다. (1)~(4)에 해당하는 level을 답하시오.

(부분점수 각 1점, 모두 맞추면 5점)



(1) tissue (2) organ (3) organ system (4) organism

2. 다음은 여성의 생식 주기를 조절하는 호르몬의 작동 순서를 요약한 표이다. 다음 빈칸에 알맞은 호르몬의 이름을 적으시오. (부분점수 각 1.5점, 모두 맞추면 5점)



(1) FSH (2) LH

(3) Estrogen (4) Progesterone : (3)과 (4)는 순서가 바뀌어도 됨.

3. 바이러스에 대한 백신이 작동하는 방식 중에 후천성 면역 (Adaptive immunity)이 있다. 후천성 면역에서 외부 침입 물질(항원)을 인식하는 방식을 2가지를 설명하시오. (부분점수 각 2점, 모두 맞추면 5점)

(1) Antigen receptor를 가지고 있는 B 세포가 Antigen을 직

접 부착하여 인식한다.

(2) Antigen receptor를 가지고 있는 T세포가 대식세포에 의해 분해된 Antigen의 부분을 인식한다.

4. 다음은 인간의 뇌의 각 부위와 기능에 대한 질문이다. 각 질문에 답하시오.

(1) 다음 표에서 각각의 부위와 기능이 일치하도록 짝을 지으시오. (부분점수 각 1점)

(1) Brainstem	(A) regulates hormones affecting many body functions, coordinates internal timekeeping
(2) Cerebellum	(B) regulates vital body functions; filters motor and sensory input; consists of midbrain, pons, and medulla oblongata
(3) Hypothalamus	(C) integrates complex information, controls personality, emotion, voluntary movement; interprets sensory information
(4) Cerebrum	(D) plans and coordinates body movements

정답: 1(B), 2(D), 3(A), 4(C)

(2) 뇌의 모양을 그림으로 그리고 위의 4개 부위의 위치를 화살표로 표시하시오. (1점)

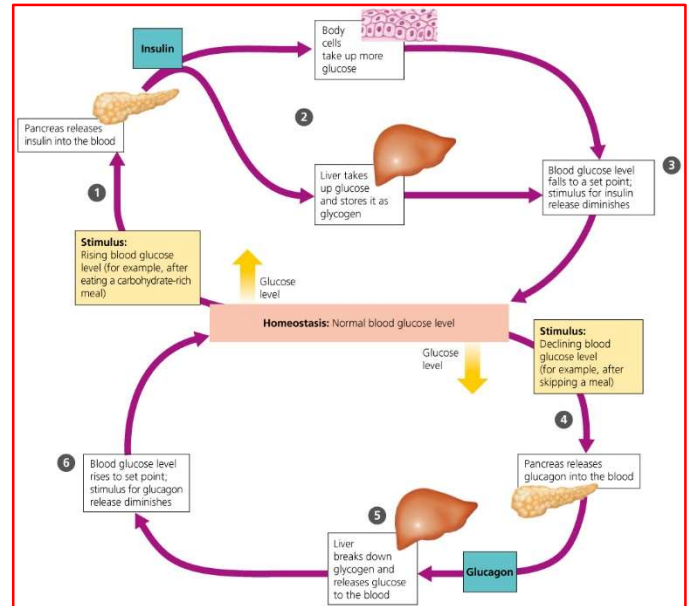
5. 동물의 진화 과정에서 가장 발달한 문(phylum)은 척삭동물(Chordate)이다. 척삭동물이 주요한 특징 4가지를 적으시오. (각 1점, 모두 맞으면 5점)

6. 지구상에 생명이 출현하게 되는 생명의 기원에 대한 가설을 4단계로 나누어 간략히 설명하시오.

7. 혈액 내의 당의 농도는 일정하게 유지되어야 한다. 당뇨병의 경우 혈액 내의 당의 농도를 조절하는 기능에 문제가 생길 경우 발생한다. 다음 각 질문에 답하시오.

(1) 혈당을 조절하는 2가지 호르몬의 이름을 제시하시오. (2점) **인슐린, 글루카곤**

(2) 위의 2가지 호르몬이 분비되는 장소와 각 호르몬의 기능을 간, 근육과 연계하여 설명하시오. (3점)



8. 종(species) 간에는 생식 장벽(reproductive barrier)이 있기 때문에 다른 종끼리는 교배가 불가능하다. 다음 질문에 답하시오.

(1) 종간에 존재하는 생식장벽의 2가지 종류를 제시하시오. (2점)

(2) 위의 (1)에서 제시한 생식장벽의 종류의 예를 각각 2가지 제시하고 설명하시오. (3점)

(1) **Prezygotic barrier, Postzygotic barrier**

(2)

A. Prezygotic barrier의 예시 중 2개

Temporal isolation, Habitat isolation, Behavioral isolation

B. Postzygotic barrier의 예시 중 2개

Reduced hybrid viability, Reduced hybrid fertility, Hybrid breakdown